

Table Schema

Table: orderdetails

Columns:

order_id	int
product_id	int
quantity	int
price_per_unit	int

Table: customers

Columns:

customer_id	int
name	text
location	text

Table: orders

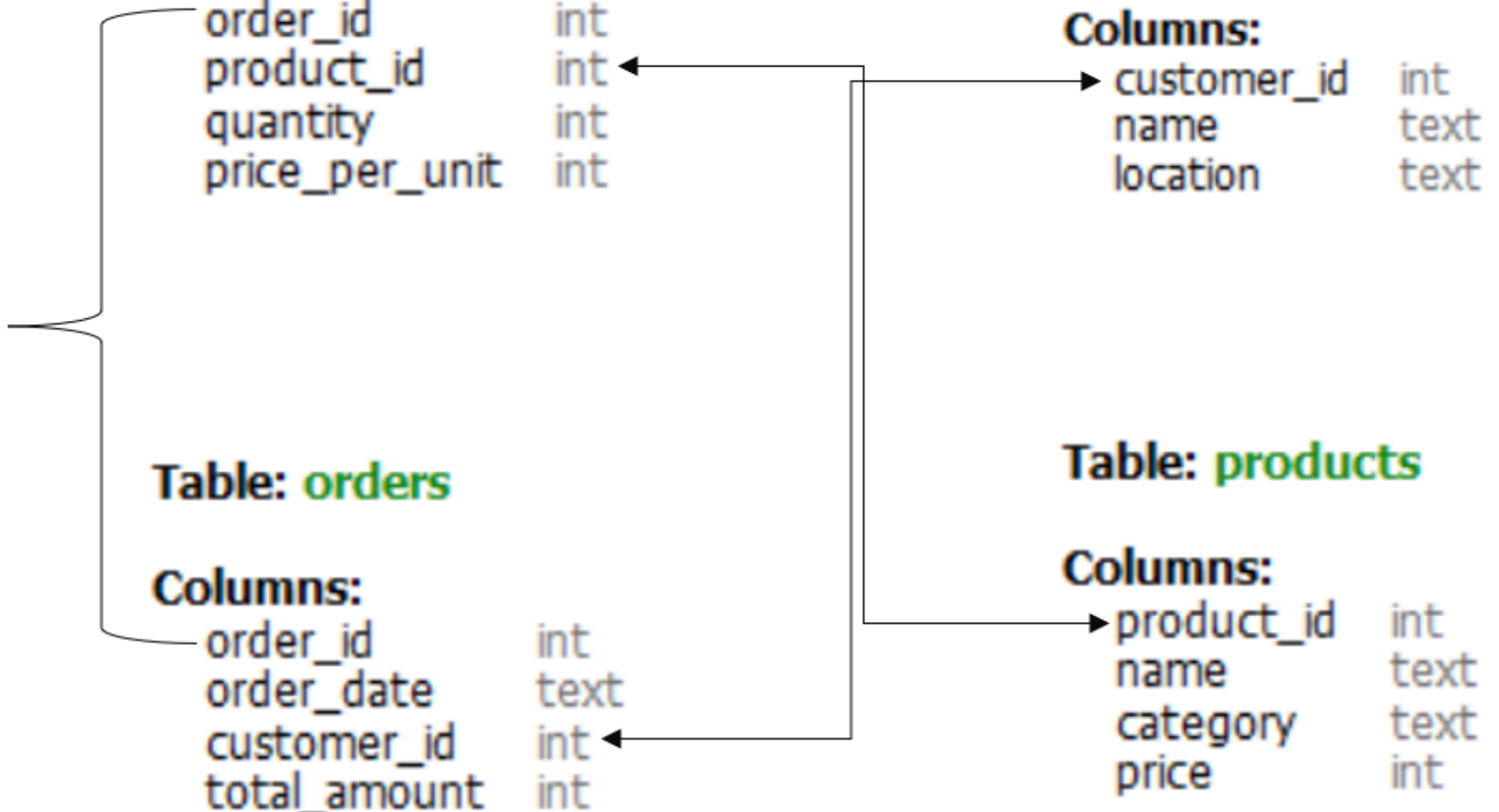
Columns:

order_id	int
order_date	text
customer_id	int
total_amount	int

Table: products

Columns:

product_id	int
name	text
category	text
price	int



Identify the top 3 cities with the highest number of customers to determine key markets for targeted marketing and logistic optimization.

```
SELECT
    location, COUNT(customer_id) AS number_of_customers
FROM
    customers
GROUP BY location
ORDER BY number_of_customers DESC
LIMIT 3;
```



This cities must be focused as a part of marketing strategies

Result Grid			Filter Rows:	
	location	number_of_customers		
▶	Delhi	16		
	Chennai	15		
	Jaipur	11		



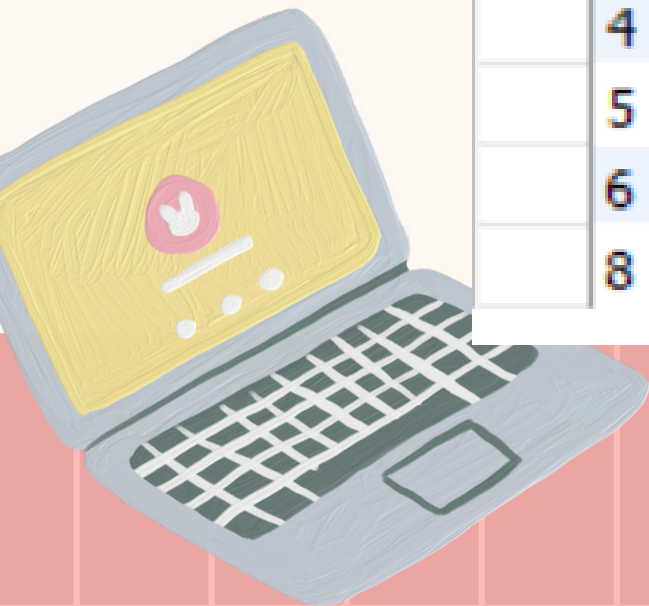
Determine the distribution of customers by the number of orders placed. This insight will help in segmenting customers into one-time buyers, occasional shoppers, and regular customers for tailored marketing strategies.

```
SELECT
    NumberOfOrders, COUNT(NumberOfOrders) AS CustomerCount
FROM
    (SELECT
        customer_id, COUNT(order_id) AS NumberOfOrders
    FROM
        Orders
    GROUP BY customer_id) AS CustomerOrders
GROUP BY NumberOfOrders
ORDER BY NumberOfOrders ASC;
```



As number of order increases the customer count decreases. In this analysis we can see occasional shoppers more.

Result Grid   Filter Rows:		
	NumberOfOrders	CustomerCount
▶	1	26
	2	26
	3	18
	4	6
	5	6
	6	1
	8	1



For each product category, calculate the unique number of customers purchasing from it. This will help understand which categories have wider appeal across the customer base.

```
SELECT
    p.category,
    COUNT(DISTINCT o.customer_id) AS unique_customers
FROM
    Products p
    JOIN
    OrderDetails od ON p.product_id = od.product_id
    JOIN
    Orders o ON od.order_id = o.order_id
GROUP BY p.category
ORDER BY unique_customers DESC;
```



Electronics category needs more focus as it is in high demand among the customers

Result Grid					Filter Rows:	
	category	unique_customers				
▶	Electronics	79				
	Wearable Tech	61				
	Photography	45				



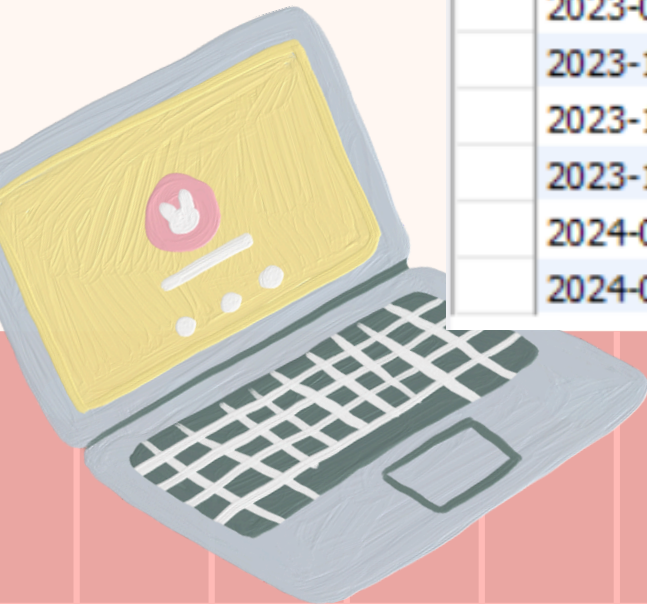
Analyze the month-on-month percentage change in total sales to identify growth trends.



```
WITH MonthlySales AS (  
    SELECT  
        DATE_FORMAT(order_date, '%Y-%m') AS Month,  
        SUM(total_amount) AS TotalSales  
    FROM Orders  
    GROUP BY DATE_FORMAT(order_date, '%Y-%m')  
)  
SELECT  
    Month,  
    TotalSales,  
    ROUND(  
        (  
            (TotalSales - LAG(TotalSales) OVER (ORDER BY Month))  
            / NULLIF(LAG(TotalSales) OVER (ORDER BY Month), 0)  
        ) * 100,  
        2  
    ) AS PercentChange  
FROM MonthlySales  
ORDER BY Month;
```

As per Sales Trend Analysis, During Feb 2024 month the sales experienced the largest decline and sale fluctuated with no clear trend.

Result Grid			
		Filter Rows:	
		Export:	
	Month	TotalSales	PercentChange
▶	2023-03	789000	NULL
	2023-04	1704000	115.97
	2023-05	1582000	-7.16
	2023-06	1040000	-34.26
	2023-07	2568000	146.92
	2023-08	1800000	-29.91
	2023-09	2927000	62.61
	2023-10	1497000	-48.86
	2023-11	1151000	-23.11
	2023-12	2774000	141.01
	2024-01	1555000	-43.94
	2024-02	396000	-74.53



Examine how the average order value changes month-on-month. Insights can guide pricing and promotional strategies to enhance order value.

```
WITH MonthlyOrderValues AS (  
    SELECT  
        DATE_FORMAT(order_date, '%Y-%m') AS Month,  
        ROUND(AVG(total_amount), 2) AS AvgOrderValue  
    FROM Orders  
    GROUP BY DATE_FORMAT(order_date, '%Y-%m')  
)  
SELECT  
    Month,  
    AvgOrderValue,  
    ROUND(  
        AvgOrderValue - LAG(AvgOrderValue) OVER (ORDER BY Month),  
        2  
    ) AS ChangeInValue  
FROM MonthlyOrderValues  
ORDER BY month;
```



December had the highest change in the average order value

Result Grid		Filter Rows:	
	Month	AvgOrderValue	ChangeInValue
▶	2023-03	60692.31	NULL
	2023-04	81142.86	20450.55
	2023-05	87888.89	6746.03
	2023-06	104000.00	16111.11
	2023-07	98769.23	-5230.77
	2023-08	112500.00	13730.77
	2023-09	121958.33	9458.33
	2023-10	83166.67	-38791.66
	2023-11	95916.67	12750.00
	2023-12	132095.24	36178.57
	2024-01	129583.33	-2511.91
	2024-02	44000.00	-85583.33



Based on sales data, identify products with the fastest turnover rates, suggesting high demand and the need for frequent restocking.

```
SELECT  
    product_id, COUNT(order_id) AS SalesFrequency  
FROM  
    OrderDetails  
GROUP BY product_id  
ORDER BY SalesFrequency DESC  
LIMIT 5;
```

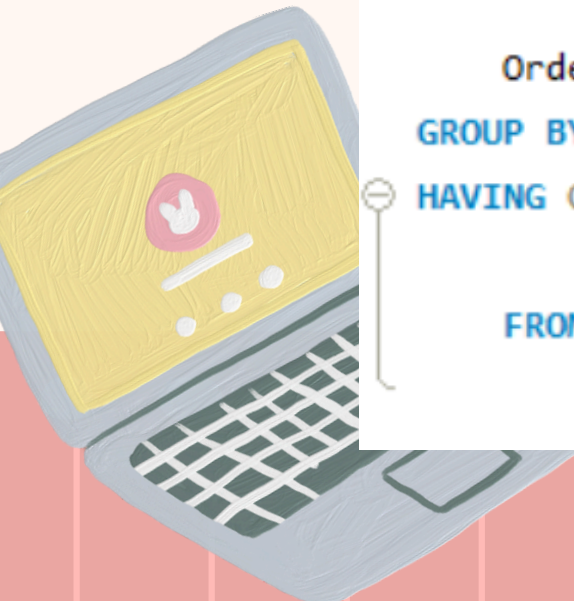


product_id 7 has the highest turnover rates and needs to be restocked frequently

Result Grid			Filter Rows:
	product_id	SalesFrequency	
▶	7	78	
	3	68	
	4	68	
	2	67	
	8	65	



List products purchased by less than 40% of the customer base, indicating potential mismatches between inventory and customer interest.



```
SELECT
    p.product_id,
    p.name,
    COUNT(DISTINCT o.customer_id) AS UniqueCustomerCount
FROM
    Products p
    JOIN
    OrderDetails od ON p.product_id = od.product_id
    JOIN
    Orders o ON od.order_id = o.order_id
GROUP BY p.product_id , p.name
HAVING COUNT(DISTINCT o.customer_id) < (SELECT
    COUNT(*)
    FROM
        Customers) * 0.40;
```

Implement targeted marketing campaigns to raise awareness and interest.

Result Grid				Filter Rows:	Ex
	product_id	name	UniqueCustomerCount		
▶	1	Smartphone 6"	36		
	8	Wireless Earbuds	38		

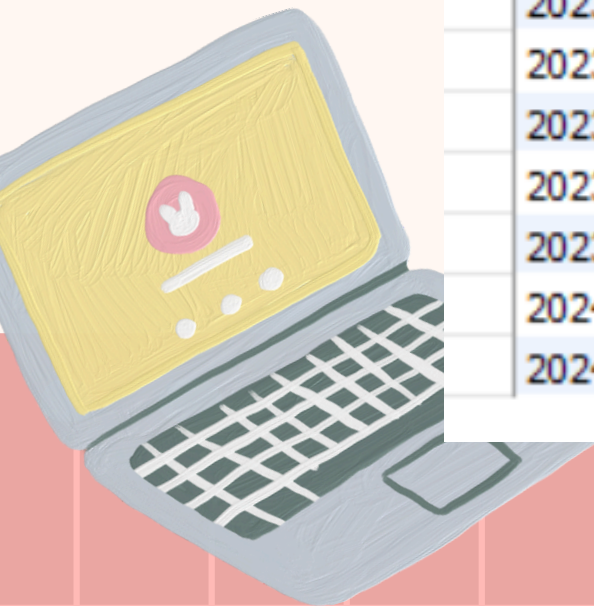


Evaluate the month-on-month growth rate in the customer base to understand the effectiveness of marketing campaigns and market expansion efforts.

```
WITH MonthlyNewCustomers AS (  
    SELECT  
        DATE_FORMAT(MIN(order_date), '%Y-%m') AS FirstPurchaseMonth  
    FROM Orders  
    GROUP BY customer_id  
)  
SELECT  
    FirstPurchaseMonth,  
    COUNT(*) AS TotalNewCustomers  
FROM MonthlyNewCustomers  
GROUP BY FirstPurchaseMonth  
ORDER BY FirstPurchaseMonth;
```

Growth Trend is downward which implies marketing campaign are not much effective.

Result Grid			Filter Rows:	Export:
	FirstPurchaseMonth	TotalNewCustomers		
▶	2023-03	11		
	2023-04	18		
	2023-05	11		
	2023-06	8		
	2023-07	11		
	2023-08	9		
	2023-09	5		
	2023-10	3		
	2023-11	1		
	2023-12	4		
	2024-01	2		
	2024-02	1		



Identify the months with the highest sales volume, aiding in planning for stock levels, marketing efforts, and staffing in anticipation of peak demand periods.

```
SELECT
    DATE_FORMAT(order_date, '%Y-%m') AS Month,
    SUM(total_amount) AS TotalSales
FROM
    Orders
GROUP BY DATE_FORMAT(order_date, '%Y-%m')
ORDER BY TotalSales DESC
LIMIT 3;
```



September, December, July month require major restocking of product and increased staffs.

Result Grid			Filter Rows:	
	Month	TotalSales		
▶	2023-09	2927000		
	2023-12	2774000		
	2023-07	2568000		

